

L02 Reflective Journal - ML Development Tools

Patrick Gandjouon Tchassem

This lab helped me understand how Python tools work together in machine learning. Before the lab, I knew about pandas, NumPy, and matplotlib, but I did not fully understand how they connected. I learned that pandas helps organize data, NumPy performs calculations, and matplotlib makes data easier to understand through visualizations.

Working with the Iris dataset showed me that looking at raw data is not always enough. The scatter plot made it easier to see that Setosa was clearly separated from the other species, while Versicolor and Virginica overlapped more. This helped me understand why visualization is important before building a machine learning model.

I also learned how statistics such as mean and standard deviation help us understand data. The lab made me think about which features, such as petal length and width, might be more useful for classification.

Overall, my biggest takeaway is that understanding and exploring data is an important part of machine learning. I also learned that clear comments, graphs, and explanations make a notebook easier to understand and review.

Anavictoria Cavazos

In this assignment, I learned more about how these tools (pandas, NumPy, and matplotlib) function differently but make important contributions when used properly. As a newcomer to this field, I now understand that each cell works differently and, according to its proper code, works for that specific cell but may not be the same for others. Also, being able to experiment with each cell helped me see how certain properties of the cell can change. For instance, in the section where there was a graph, I played with the colors a little, which the cell was able to run with the change I had made. However, when I tried to change other aspects, such as writing the text for the field 'show()', I noticed that only certain words were accepted, so when I typed words that were not specific, it would not run the cell successfully. Going deeper into the learning aspect of this Lab, I made the observation that each cell that contained a visualization of the datasets was different in spite of containing the same three classes. There was a bar chart and a scatter plot, both containing the same components of species, but showing different aspects of the data. This was able to be seen visually, but was seen in detail with the code

shown in the cell. The breakdown of it shown through the code helped me see the wording of the codes and what each represented according to the iris dataset plot. In addition, seeing the sepal length, sepal width, petal length, and petal width in code with exact numbers shows how precision or accuracy of the data is harder to tell from a visualization as opposed to a specific dataset code. To summarize although at first I was confused about how to start the assignment even after reading the instructions and reviewing the cells, I did not quite catch the idea or purpose or what had to be done. However, as I went through each cell and was able to properly do each one, this reflective journal is what I now see helped me develop my true understanding of practicing and analyzing the datasets provided for lab 02.

Hashim Sayed Hoosini

For this project, I used Google Colab and installed required libraries, such as Pandas, Matplotlib for charts and graphs, and Scikit-learn, which helped me learn different tools in the library used in machine learning. It showed me how they are connected and work together. I learned Pandas organizes the data, NumPy does the calculations, and Matplotlib turns the data into charts and graphs. Then I loaded the Iris Flower dataset. I worked through all the cells in the lab and ensured each cell ran and worked properly by reviewing the output. I analyzed the dataset by looking at features, samples, and classes. I observed 150 samples, 4 features, and 3 classes are used. My key finding was Setosa was clearly separated from the other two species. Versicolor and Virginica overlapped in the scatter plot. Virginica generally has longer sepals than the other species. Overall, this project provided me with a better understanding of tools and libraries in machine learning. I learned it is important to explore and understand the data and turn the data into visualization charts for better understanding. LLM used: ChatGPT Propose: spelling check and grammar error

Kenneth Kouokam

My main takeaway from this lab is that understanding data matters more than simply getting code to run. Pandas, NumPy, and matplotlib each help answer a different question: how the information is organized, what its numbers tell us, and what patterns are visible. I see the value of using these tools together because a table, a calculation, and a chart can explain different parts of the same dataset. This makes machine learning preparation feel more concrete: useful predictions begin with understanding the information available. One distinction I need to keep clear is the difference between a feature and a label. The Iris table has five columns, but only four are measurement features. The species column contains the category that a future model

would try to predict. Thinking of the table as a spreadsheet makes the structure easier to understand, but the purpose of each column matters more than its position. This distinction helps me connect ordinary data organization with the way machine learning problems are defined. The numerical results also show why one summary is not enough. The average sepal length is about 5.84 cm, while the standard deviation is about 0.83 cm. I interpret the first number as the center of the measurements and the second as a measure of their spread. Neither number describes every flower. Comparing the averages by species makes the differences clearer, but even those averages cannot show all the variation within each group. This reminds me to ask what a calculation leaves out before drawing a conclusion. The scatter plot provides the clearest example of that limitation. Setosa tends to form a relatively distinct group, while versicolor and virginica overlap considerably. I can see why a difference between group averages does not guarantee that individual flowers are easy to classify. The bar chart adds another perspective: all three species have 50 examples. Equal counts help describe the dataset, but they do not prove that a model would perform well. A useful next question is whether petal measurements would separate the species more clearly. Another lesson I take from the notebook is that checking the work is part of the analysis. There are no missing values here, so filling missing data would be unnecessary. Running cells in order also matters because later steps depend on earlier variables. My approach to future labs should be to inspect the data, explain the outputs in plain language, and then check that the notebook runs from the beginning. Being able to explain why a result makes sense is a stronger sign of understanding than a success message alone.

Saimi Manasiya

Reflection This project helped me understand the basics of data analysis and machine learning. I learned how to use Google Colab, Python, Pandas, NumPy, and Matplotlib to work with and visualize data. I also learned how important it is to carefully follow the instructions, check the results, and understand what the code is doing. Overall, this project gave me more confidence in working with Python and data analysis.